

Response letter

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Outline

In their letter, Bogdziewicz et al. say that the reproduction synchrony among individuals has to be controlled by the solstice. We do not disagree with this idea—as it is explicitly one of the hypotheses our paper considers, and we do not believe our results invalidate it. While our study does not address the risk of "late frost damage", as mentioned by Bogdziewicz et al., we appreciate their interest in our work. Our results do show solstice is optimal, but also highlight a certain amount of variation across space.

The notion of "temperature sensing window" (in fact, some amount on correlation with a rolling window) appears higher around the solstice (and may even start earlier, as shown in their Fig1). Interestingly, the timing of this correlation between temperature and seed production also appear to vary across space (Journe et al., Fig. 1 in Bogdziewicz). However, the lack of quantification of this spatial variation—alongside an omission of possible temporal variation—make it difficult to compare their findings across latitude and with our results. Clearly, variation exists and will happen, but the extent to which this variation invalidate the "celestial starting gun" hypothesis remains unclear. Similarly, we believe that the lack of a true quantification of the scale under which the "remarkable spatial synchrony" is happening is an important missing piece of this hypothesis.